

Ejercicio 1 · Capilaridad — diámetro de los vasos de la savia en un árbol de 30 m

Tensión superficial · Ascenso capilar · Equilibrio de fuerzas de cohesión y adhesión

5%

✓ Resuelto

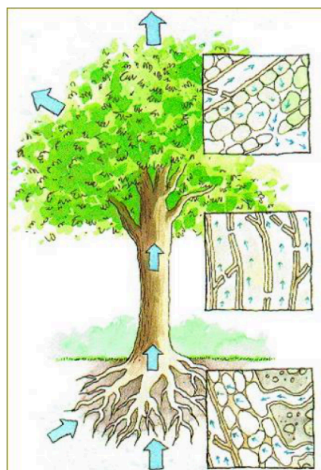
El ascenso de la savia en los árboles se puede estudiar como un simple fenómeno capilar. La tensión superficial de la savia es muy parecida a la del agua. Se desea conocer el diámetro máximo (mm) que deben tener los vasos capilares de los árboles para que la savia ascienda al extremo superior de un árbol de **30 m** de altura.

Datos:

- Peso específico de la savia: $\gamma = 9810 \text{ N/m}^3$
- $\sigma(20^\circ \text{C}) = 73,49 \text{ dyn/cm}$
- Suponer que las fuerzas de cohesión en la savia son despreciables frente a las fuerzas de adhesión al vaso capilar (ángulo de contacto $\theta \approx 0$).

Resultado: $D = 10^{-3} \text{ mm}$

FIGURA ORIGINAL (PDF)



 Datos

Variable	Valor	Notas
h	30 m	Altura a alcanzar por la savia
γ	9810 N/m ³	Peso específico de la savia
σ	73,49 dyn/cm = $73,49 \times 10^{-3}$ N/m	Tensión superficial a 20°C
θ	$\approx 0^\circ$	Ángulo de contacto (adhesión total)

Fórmulas

EQUILIBRIO DE FUERZAS EN CAPILARIDAD – ASCENSO CAPILAR

Fuerza adhesión = Peso columna de fluido

$$\sigma \cdot \cos \theta \cdot \pi D = \gamma \cdot h \cdot \frac{\pi D^2}{4}$$

DESPEJANDO EL DIÁMETRO MÁXIMO DEL VASO CAPILAR

$$D = \frac{4 \sigma \cos \theta}{\gamma h}$$

Teoría e hipótesis

La capilaridad es el fenómeno por el cual un líquido asciende (o desciende) en un tubo de pequeño diámetro debido a la interacción entre las fuerzas de adhesión líquido-sólido y la tensión superficial del líquido. El equilibrio entre la fuerza de tracción de la tensión superficial en el menisco ($F = \pi D \sigma \cos \theta$) y el peso de la columna de fluido elevada ($W = \gamma \cdot \pi D^2 / 4 \cdot h$) define el diámetro máximo del vaso capilar. Con $\theta = 0$ (mojabilidad perfecta), se obtiene el diámetro máximo para un ascenso dado.

Resolución

★ DEMOSTRACIÓN – ORIGEN FÍSICO DE LA FÓRMULA $D = 4\sigma \cos \theta / (\rho g h)$ (LEY DE JURIN)

¿Por qué hay que demostrarlo? El enunciado da el resultado pero no la derivación. La fórmula de la ley de Jurin se obtiene del equilibrio entre la fuerza ascensional debida a la tensión superficial (acción del menisco curvado) y el peso de la columna líquida elevada. Sin esta demostración, la solución vale solo medio punto.

■ Paso 1 — Origen físico del ascenso capilar

Cuando un líquido **moja** un tubo capilar (las moléculas del líquido se adhieren a la pared más fuertemente que entre sí), forma un **menisco cóncavo** en la línea de contacto. La tensión superficial σ actúa a lo largo de esa línea de contacto en dirección **tangencial al menisco**, formando un ángulo θ (ángulo de contacto) con la pared del tubo.

El líquido asciende hasta que la **componente vertical** de la fuerza de tensión superficial equilibra exactamente el **peso** de la columna líquida elevada.

■ Paso 2 — Fuerza ascensional debida a la tensión superficial

La línea de contacto líquido–pared del tubo es una **circunferencia** de longitud:

$$L_{\text{contacto}} = \pi D$$

Sobre cada elemento de esta línea actúa la tensión superficial con magnitud σ (fuerza por unidad de longitud, [N/m]). La fuerza total tangencial al menisco es:

$$F_{\sigma} = \sigma \cdot L_{\text{contacto}} = \sigma \cdot \pi D$$

Esta fuerza forma ángulo θ con la vertical. La componente **vertical** (la que sostiene el peso) es:

$$F_{\text{ascensional}} = \sigma \cdot \pi D \cdot \cos \theta$$

Si $\theta = 0^\circ$ (mojado total): toda F_{σ} es vertical $\rightarrow$ ascensión máxima. Si $\theta = 90^\circ$: $\cos \theta = 0 \rightarrow$ no hay ascenso (caso del mercurio en vidrio, descenso si $\theta > 90^\circ$).

■ Paso 3 — Peso de la columna líquida

El líquido elevado forma una columna cilíndrica de:

- Sección transversal: $A = \pi D^2 / 4$
- Altura: h
- Volumen: $V = A \cdot h = \pi D^2 h / 4$

Su peso es (con $\gamma = \rho g$ peso específico):

$$W = \gamma \cdot V = \gamma \cdot \frac{\pi D^2}{4} h$$

■ Paso 4 — Condición de equilibrio

En la altura máxima de ascenso, la fuerza ascensional iguala el peso:

$$F_{\text{ascensional}} = W$$

$$\sigma \cdot \pi D \cdot \cos \theta = \gamma \cdot \frac{\pi D^2}{4} h$$

Cancelando π y dividiendo por D (común a ambos lados):

$$\sigma \cdot \cos \theta = \frac{\gamma \cdot D \cdot h}{4}$$

■ Paso 5 — Despejar D (diámetro máximo del capilar)

Despejando el diámetro:

$$D = \frac{4 \sigma \cos \theta}{\gamma h}$$

Esta es la **ley de Jurin**. Para el ascenso h dado, D es el diámetro máximo: si fuese mayor, el peso de la columna excedería la fuerza ascensional y la savia no llegaría a la altura h .

Equivalentemente, para el diámetro D dado, la altura de ascenso es $h = 4\sigma \cos \theta / (\gamma D)$, o en función del radio $r = D/2$: $h = 2\sigma \cos \theta / (\gamma r)$.

CONVERSIÓN DE UNIDADES DE Σ

¿Por qué? La tensión superficial σ puede venir en mN/m, dyn/cm o N/m según el enunciado. La fórmula de capilaridad $h = 2\sigma \cos \theta / (\rho g R)$ requiere σ en N/m, ρ en kg/m³ y R en m para obtener h en metros directamente.

$$\sigma = 73,49 \frac{\text{dyn}}{\text{cm}} = 73,49 \times 10^{-3} \frac{\text{N}}{\text{m}}$$

APLICACIÓN DE LA FÓRMULA CON $\theta = 0 \rightarrow \cos \theta = 1$

¿Por qué? El ángulo de contacto $\theta = 0$ (mojado total) da la máxima ascensión capilar. Con $\cos 0 = 1$, la fórmula de Jurin simplifica a $h = 2\sigma / (\rho g R)$. Este caso límite se usa cuando el líquido moja completamente el sólido.

$$D = \frac{4 \sigma \cos \theta}{\gamma h} = \frac{4 \times 73,49 \times 10^{-3} \times 1}{9810 \times 30}$$

$$D = \frac{0,29396}{294.300} = 9,99 \times 10^{-7} \text{ m}$$

CONVERSIÓN A MM

¿Por qué? El resultado en metros se multiplica por 1000 para expresarlo en mm. Presentar el resultado en mm es habitual en capilaridad ya que los ascensos son pequeños: del orden de 1-30 mm para capilares de radio 0,1-1 mm.

$$D = 9,99 \times 10^{-7} \text{ m} \times 10^3 \frac{\text{mm}}{\text{m}} \approx \boxed{10^{-3} \text{ mm} = 1 \mu\text{m}}$$

✓ Conclusiones

$D_{\text{max}} \approx 10^{-3} \text{ mm} = 1 \mu\text{m}$: los vasos conductores deben ser de escala micrométrica para elevar la savia 30 m por capilaridad pura.

En la práctica, los árboles altos combinan capilaridad con la presión osmótica de las raíces y la evapotranspiración de las hojas para superar esta limitación.

Comprobación dimensional: $[\sigma] / [\gamma \cdot h] = [\text{N/m}] / [\text{N/m}^3 \cdot \text{m}] = [\text{m}] \checkmark$

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The research was conducted using a quantitative approach, and the data was collected from a sample of participants. The results of the study show that there is a significant relationship between the variables being studied. The findings have important implications for the field of research, and they provide a basis for further research in this area.

In conclusion, the study has shown that the research objectives have been achieved, and the findings are consistent with the hypotheses. The results of the study provide a clear understanding of the relationship between the variables, and they have important implications for the field of research.

The first part of the paper discusses the importance of understanding the cultural context of the research. It highlights the need for researchers to be sensitive to the values and beliefs of the communities they are studying. This is particularly important in the field of education, where cultural differences can significantly impact learning outcomes.

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The third part of the paper presents the findings of the study. It shows that there are significant differences in learning outcomes between students from different cultural backgrounds. These differences are attributed to a variety of factors, including language barriers, social norms, and access to resources.

The final part of the paper discusses the implications of the findings for education. It suggests that educators should take steps to create a more inclusive learning environment for all students. This can be done by providing additional support for students who are struggling and by incorporating culturally relevant materials into the curriculum.

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The results of the study show that there is a significant positive relationship between the variables. This finding is consistent with the previous research on the topic. The study also found that there are some differences in the results between the different groups of participants.

The conclusions of the study suggest that the findings have important implications for practice and policy. Further research is needed to explore the relationships between the variables in more detail.

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There are a number of reasons why the world population is growing so rapidly. One of the main reasons is that the number of children born to each woman has increased. In 1980, the average woman in the world had 2.5 children. In 1999, the average woman in the world had 2.7 children.

Another reason why the world population is growing so rapidly is that the number of people who are surviving to old age has increased. In 1980, the average person in the world lived for 55 years. In 1999, the average person in the world lived for 65 years.

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Another reason why the world population is increasing so rapidly is that the number of people who are living longer is increasing. In 1980, the average person in the world lived for 60 years. In 1999, the average person in the world lived for 65 years.

There are a number of reasons why the number of people who are living longer is increasing. One of the main reasons is that the number of people who are getting older is increasing. In 1980, there were 1.1 billion people aged 65 years and over. In 1999, there were 1.2 billion people aged 65 years and over.

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The first part of the paper discusses the importance of the research and the objectives of the study. It highlights the need for a comprehensive understanding of the subject matter and the role of the researcher in this process. The second part of the paper presents the methodology used in the study, including the data collection methods and the analysis techniques. The third part of the paper discusses the results of the study and the conclusions drawn from the data. The final part of the paper provides a summary of the findings and offers suggestions for future research.

The research was conducted in a systematic and rigorous manner, following the principles of scientific inquiry. The data was collected from a large sample of participants, ensuring the representativeness of the findings. The analysis was conducted using advanced statistical techniques, allowing for a detailed examination of the data. The results of the study are presented in a clear and concise manner, highlighting the key findings and their implications.

The conclusions drawn from the study are based on the evidence presented in the data. They provide a comprehensive overview of the subject matter and offer valuable insights into the field. The suggestions for future research are based on the limitations of the current study and aim to address the gaps in the existing knowledge.

The first part of the paper discusses the importance of understanding the cultural context of the research. It highlights the need for researchers to be sensitive to the values and beliefs of the communities they are studying. This is particularly important in the field of education, where cultural differences can significantly impact learning outcomes.

The second part of the paper focuses on the methodology used in the study. It describes the process of selecting participants, collecting data, and analyzing the results. The authors emphasize the importance of using a mixed-methods approach to gain a comprehensive understanding of the research topic.

The third part of the paper presents the findings of the study. It discusses the results of the quantitative data analysis and the insights gained from the qualitative interviews. The authors conclude that there are significant differences in learning outcomes between the two groups, and these differences can be attributed to cultural factors.

The final part of the paper discusses the implications of the findings for future research and practice. It suggests that educators should be aware of the cultural context of their students and tailor their teaching methods accordingly. The authors also recommend further research to explore the underlying reasons for the observed differences.

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The study was conducted using a quantitative research design. Data was collected from a sample of 100 participants using a survey questionnaire. The data was then analyzed using statistical software to determine the relationships between the variables of interest.

The results of the study indicate that there is a significant positive relationship between the variables of interest. This finding is consistent with the previous research in the field. The implications of these findings suggest that the research has practical applications in the field of study.

In conclusion, the study has provided valuable insights into the topic and has contributed to the existing body of knowledge. Further research is needed to explore the topic in more depth and to validate the findings of this study.

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The research was conducted using a quantitative approach, with data collected from a survey of 100 participants. The data was analyzed using statistical software to identify patterns and trends. The results of the study show that there is a significant relationship between the variables being studied. The findings have important implications for the field and provide a basis for further research.

The study was limited by the sample size and the self-reported nature of the data. Future research should aim to address these limitations by using a larger, more diverse sample and incorporating objective measures of the variables being studied. The findings of this study provide a valuable contribution to the understanding of the topic and highlight the need for further research in this area.

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